



Zero Day Engineering

research & intelligence

ZERO DAY VULNERABILITY RESEARCH

TRAINING OVERVIEW

This 4-day training course comprehends essential knowledge and skills required to conduct modern low-level security research, vulnerability exploit development, secure software engineering and hardening. The course is created and taught by a professional vulnerability researcher and low-level hacker who is actively engaged in security bounty hunting and zero day vulnerability discovery and exploitation as her main job, and qualified as a top zero day hacker by successful participation in Pwn2Own competitions.

This course is designed for complete beginners, yet inclusive of many advanced aspects of the craft that students will likely need as they advance. Following the always systematical structure of Zero Day Engineering Project courses, it covers all the essentials that students need to get started with arbitrary research targets in a compact and practical way. Excluded are specialized topics: advanced exploit techniques and mitigations, OS & hardware theory, and threat models for specific software classes.

Training course theory will be illustrated and practiced with modern and immediately relevant real-life examples, including high-impact vulnerability case studies and non-trivial examples.

AUDIENCE

This training course is primarily intended for beginners in low-level security research with basic knowledge in computer science.

Materials of this course comprehend an essential base of knowledge and skills for software vulnerability researchers, security-minded software developers, and DevSecOps engineers.

Systematical structure and strategic insights of this course would be valuable for seasoned hackers and professional security researchers.

PREREQUISITES

Mandatory:

- General computer science.
- A laptop with a modern Linux VM.

Optional:

- C-language.
- Familiarity with basic concepts of low-level computer security.

OBJECTIVES

Upon completion of this training course the students should – with a certain commitment – be able to:

- Engage in bug bounty hunting.
- Participate in CTF competitions.
- Conduct exploit & vulnerability analysis.
- Find simple oday vulnerabilities.
- Reverse-engineer simple softwares..
- Avoid security bugs in their code.
- Write and customize fuzzing tools.
- Apply public tools and scripts in the right order.
- Make independent and reasonably correct judgements about 3rd party bugs & exploits.
- Choose further specialization and research focus wisely.

LEVELS & CERTIFICATION

Complexity: beginners, involved.

Certificate grade: CoDEo (Foundation).

Next level training: "<Target> Vulnerability Research"

TRAINING DETAILS

This training course assumes an extremely ambitious goal: to comprehend everything you need to get started with modern application security research in just 4 days. This is accomplished by following a deeply systematical learning model created by the instructor with an extensive experience in multiple specializations of the subject, alongside with time-proven techniques of cognitive optimization.

Despite the sassy name, the knowledge presented in this course is ethically neutral. It provides a universal foundation theory and skills for writing secure code, as much as for offensive security research of arbitrary software, firmware and hardware targets. Students are encouraged to think generically in terms of universal models, that makes a solid ground for arbitrary specializations: such as bug hunting, writing fuzzing tools, exploit development, reverse-engineering, or malware analysis.

The course deliberately ignores any differences between attacking binary code and web applications, because the underlying threat models are essentially the same, as explained on Day 1. That said, course approach is heavily geared towards binary analysis. Even logic vulnerabilities – the bug class which is commonly associated with web applications – are considered with respect to modern binary programs.

Key principle is to expose attendees immediately to some of the most impactful real-life case studies. While it may be impossible to learn it all in a few days, eager students are expected to be able to enter CTF competitions and find some easy bugs straight out of the training. In addition, the solid foundation given by this course will later help them to learn specialized knowledge faster and more systematically.

An empirical sampling of various specializations in the field of application security, alongside with insights of a seasoned hacker, comes as a side effect in this course. That would enable newcomers to choose their specialization based on personal experience rather than theoretical reasoning or fashion.

Course experience is a mixture of theory and practice. Practicals – that's small exercises and labs – are based on Linux, as the most versatile technical research platform that every low-level hacker eventually employs extensively in their workflow, even if their research is focused on a different OS.

The knowledge presented in this course is 100% original, based on independent technical research and reverse engineering work done by the author herself, and illustrated with security vulnerabilities discovered by the author, as well as with publicly available sources. Course content is kept up-to-date.



ABOUT THE INSTRUCTOR

Alisa Esage is an independent hacker and founder of Zero Day Engineering. She was the first woman to compete solo at Pwn2Own, where she demonstrated a critical virtual machine escape. She currently works at the cutting edges of the exploit field, between advanced theoretical research and non-trivial exploit engineering for traditionally challenging research targets; such as hypervisors, browsers and basebands.

Alisa is known for her ability to reverse engineer structural patterns behind some of the most mysterious zero-days and architectures, and compress them into practical, high-impact methodologies. Her trainings are used by top-tier researchers, government analysts, and professional red teams worldwide.

PROGRAM AT A GLANCE

DAY 1. FOUNDATION

Abstract models and essential general theory.

Theory (2 h)

- Mindset.

What makes an effective zero day engineer?

- Program theory concepts.

System design, state space, code correctness.

Core topic (4 h)

- Model of Vulnerability.

Bug vs. Vuln. Root cause vs. effect. Exploitability

- Model of Exploit.

Objectives, stages, primitives, payload.

- Application Threat Models.

Attack surfaces & attack vectors in applications.

Essential skills & research topics (2 h)

- General vulnerability classification.

Design bugs, logic bugs, memory safety bugs.

- C and C++, JavaScript, Python.

Overview & exploitation-relevant bits.

DAY 3. BINARY HACKING

Get started with binary code analysis and exploit development.

Theory (2 h)

- Compiler theory.

From HLL to binary code.

- Assembly programming vs. shellcoding.

CPU models & popular architectures.

Core topic (4 h)

- Fuzzing essentials: basic & next level.

From random inputs to vulnerability discovery.

Essential skills & research topics (2 h)

- Memory safety bugs vs. exploit mitigations.

Classification & common patterns. Case studies.

- Reverse code engineering.

From binary to assembly to code to architecture.

- Binary debugging.

DAY 2. PROGRAM ANALYSIS & LOGIC BUGS

Working with source code. Discovering, evaluating and reproducing simple types of bugs.

Theory (2 h)

- Logic bugs.

Classification and common patterns.

- Case studies.

High-impact logic bugs in modern software.

Core topic (4 h)

- Static code analysis.

Bug hunting in large and complex code bases.

- Assisted code analysis.

Finding bugs in code with modern tools. CodeQL.

Essential skills & research topics (2 h)

- Security patch analysis.

Understanding bugs via source code patches.

- Introduction to dynamic analysis.

Debuggers in source mode. Frida.

DAY 4. INTO THE FUTURE

Up-and-coming essentials in low-level security.

Theory (2 h)

- Mathematics of low-level security.

Program modeling efforts, weird machines, abstract interpretation.

- Memory-safe programming languages.

Rust. Internals, threats, bugs, academic research.

Core topic (4 h)

- Blockchain security.

Novel threat models. Case studies.

Research topics (2 h)

- Hardware bugs.

Speculative execution, Rowhammer, FI.

- DSP and NPU.

Software, firmware and hardware that powers artificial neural networks and deep learning.

TRAINING PACKAGES & FEES

This training course is publicly available online (live), in self-paced packages, and privately.

Online and on-demand trainings are based on a modern streaming platform with high-quality audio and video, and a group chat. The instructor will be available for questions, feedback and technical support, according to specific package conditions.

Payments through the website are subject to merchant fees, which can be waived by paying via bank transfer or with crypto currencies.

LIVE TRAINING (NOT AVAILABLE)

View-only access

What's included:

- Access to public online training.
- Guaranteed instructor's feedback by email.
- Training slides and materials.
- Training completion certificate.

Price: -,- NET per person.

Interactive access

What's included:

- Everything in the View-only option.
- Personal feedback and technical support from the instructor during the live training.
- A possibility to receive a Training Achievement certificate by undergoing an assessment.

Price: -,- NET per person.

Note: Limited number of seats for the Interactive access option.

SELF-PACED COURSE

Basic package

What's included:

- Video lectures, exercises, and walk-through.
- Training slides and supporting materials.
- Access to videos for 3 months.

Price: €4,000.- per person.

Complete package

What's included:

- Everything in the Basic package.
- One month of technical support by email.
- One on-demand consultation call with the founder.
- Training completion certificate.

Price: €6,000.- per person.

Note: certificates will be issued upon request.

PRIVATE & CUSTOMIZED TRAINING

Minimal private group size is 10 persons. Availability is limited; inquire at least 3 months in advance.

LIVE TRAINING DATES & BOOKING

Refer to our [website](#) for the dates of the nearest public training.

All our online training courses are offered exclusively at the Zero Day Engineering company website: <http://zerodayengineering.com>.

Bookings of public training seats are accepted via the website. To discuss a private training, [email us](#).

INDUSTRY FEEDBACK - ZERO DAY ENGINEERING TRAININGS



Jael Koh · 1st
OSEE | OSCE3 | ZDE VR | Corelan Advanced | WISE |
HackSys AKE
7mo · Edited · 🌟

I'm excited to start off 2024 by completing [Zero Day Engineering](#)'s Zero Day Vulnerability Research course.

[Alisa Esage](#) does an amazing job of taking something as complex and intimidating as Vulnerability Research and breaking it down into a comprehensive yet approachable curriculum.

One of my biggest takeaways from the course was Alisa's methodology for finding zero days in modern software. She teaches a systematic approach, using models of vulnerabilities, exploits, and application threats.

For example, by modeling the subsystems and threats of an application, I learned how to prioritize which subsystems to target and discover new conceptual attack vectors.

This course was both enjoyable and incredibly informative. I felt significantly better prepared for vulnerability research by the end, and I believe this systematic approach will help me avoid common pitfalls.

Overall, the course offers a powerful, systematized methodology that I haven't encountered before. It's also a great foundation for beginners interested in entering the field of Vulnerability Research.

I thoroughly enjoyed this course and am eagerly looking forward to its major upgrade later this year.



Peleg Hadar 🐦
@peleghd

The Hypervisor Exploitation One training by [@alisaesage](#) was simply incredible. Alisa is a great teacher with a lot of experience in the Vuln research field.

The Training contains practical exercises, it provides the knowledge you'll need for starting a Hypervisor vuln research!



Mohammad Hussam Alzeyyat · 2nd
Vulnerability Researcher/Cyber Security (Instructor, Labs Developer, Upwork Top Rated)/OSCE, OSMR
2mo · Edited · 🌟

I just finished the "Masterclass: Hacking open source fuzzers for smarter bughunting" from [zerodayengineering.com](#) by Alisa Esage.

The 4-hour training is an eye-opening to more advanced fuzzing and it's more oriented to learn how to customize the fuzzer for your own purpose and not to learn how to fuzz.

The masterclass focuses on AFL and WinAFL since those are the most powerful frameworks for fuzzing and explains the anatomy of fuzzing and the differences between the old generation fuzzers methodology and the new ones.

I don't think the masterclass is for beginners as much as for intermediate people or those with fuzzing experience.



Josh Pitts 🐦
@ausernamedjosh

The JavaScript Engines for Hackers by [@zerodaytraining](#) was great. Shortened the time to get familiar with V8. Thx [@alisaesage](#)!



Gal Z 🐦
@0xgalz

What a blast! Just finished [@alisaesage](#) 3 days deep technical Hypervisor Vulnerability Research course. Incredible and well designed with high quality materials.
Thanks for everything! Now, off to look at some VMM code \o/



Carlomagno A. · 2nd
Security Researcher
6mo · 🌟

My review for [#ZeroDayVulnerability](#) course by [Alisa Esage](#).

Alisa's does an amazing job at explaining more than decade of wisdom, insights and knowledge on a complex topic as Vulnerability Research in a 4 days course. The value of this information and training goes beyond just specific attack vectors or simplistic how-to's, it covers deep technical hands on training on real world vulnerable applications.

The segment of the course that I enjoyed the most was the establishment of the correct/appropriate mindset, definition of abstract models/frameworks, cognitive skills needed to better improve, this specific points matched with a well structured self-study roadmap will allow you to track your progress by establishing a feedback loop on the concepts that you need to improve upon to excel at Vulnerability Research.



rui 🐦
@fdiskyou

Alisa's training completely changed my mindset and the way I was looking at hypervisors. I highly recommend the Advanced Hypervisor Exploitation training. Actually, if you're interested in hypervisors it's mandatory.



Kapil Khot · 1st
Security Consultant/Penetration Testing | CoreLAN Exploit Dev | OSCP
7mo · Edited · 🌟

Here's my review for [#ZeroDayVulnerabilityResearch](#) class offered by [Zero Day Engineering](#). It's an intensive four days training where [Alisa Esage](#) shares some important tips from her Pwn2Own competition and decade long zero day research experience.

She walked us through real world application's code base to demonstrate how to identify its subsystems and which one to prioritize for testing. Additionally, she walked us through some of the real world vulnerable code snippets from various enterprise and real-world applications to demonstrate several bug classes such as memory corruption vulnerabilities, hardware bugs, and logic bugs etc. Furthermore, she also touched base on the security weaknesses of one of the memory safe languages and shared research tips on upcoming technologies.
One of the modules, Fuzzing Masterclass took us through the internals of coverage guided fuzzers and offered guidance on customizing them for applications not supported out of the box.

Overall, this course helps you build a zero day research mindset, guides you on what needs to be done, and forces you to figure out the most of the "how-to" part, which in my opinion is the best thing.



Mario Romero Serrano · 2nd
Senior Vulnerability Researcher - Cryptography Research
Centre at Technology Innovation Institute (TII) | Cryptograph...
5mo · 🌟

The last few months I have been enjoying and learning about vulnerability research from one of the best trainings available today.

The 4-day Zero Day Vulnerability Research course from [Zero Day Engineering](#) by [Alisa Esage](#) is not only exceptional on a technical level, but it also teaches you a methodology to follow, and personally, what for me is most valuable and concentrates Alisa's years of experience and knowledge: it introduces you to the mindset necessary for your vulnerability research to be successful.

I am already applying the knowledge I have acquired and I am eager to continue learning from her. I'm sure I'll be taking some more of her courses. For now, I will enjoy the course for the second time.

Don't miss the opportunity to do it:



Jacon Walker 🐦
@call_eax

Big thanks to [@alisaesage](#) and [@zerodaytraining](#) for their top-notch training content! Just wrapped up a refreshing course with promising results - from bug discovery to exploitable vulnerability in just 9 days. Dive into my chronicle here: <https://t.co/PCfqzCIs6B>



Veer Singh · 1st
Security Practitioner
7mo · 🌟

I recently (2023) completed the "Hypervisor Vulnerability Research" course by [Alisa Esage](#), which provided a comprehensive exploration of virtualization technology and its security aspects. The course began with a clear introduction to virtualization technology, distinguishing between various types of hypervisors. We then explored the rich functionality of guest services, understanding their practical implementation and potential security challenges. The module on attacking guest services equipped us with valuable perspective on attack vectors, and the in-depth look at VirtualBox's implementation was insightful. Overall, this course is highly recommended for both beginners and experienced professionals looking to enhance their virtualization security skills. Here is a link to the course for anyone interested - <https://lnkd.in/g5N2gNmT>

Training: Hypervisor Vulnerability Research

[zerodayengineering.com](#)



scryh 🐦
@scryh_

Just finished the [@zerodaytraining](#) hypervisor vulnerability research offline course by [@alisaesage](#). Impressive research bundled into very well structured lessons, which are communicated in a clear and comprehensible way. Learned a lot. Thank you very much!
<https://t.co/iw3UAGBUuq>

APPLICATIONS & INQUIRIES

E-mail: contact@zerodayengineering.com.

Public cohorts and booking: zerodayengineering.com

Socials: [@zerodayalpha](#) [@zerodaytraining](#)